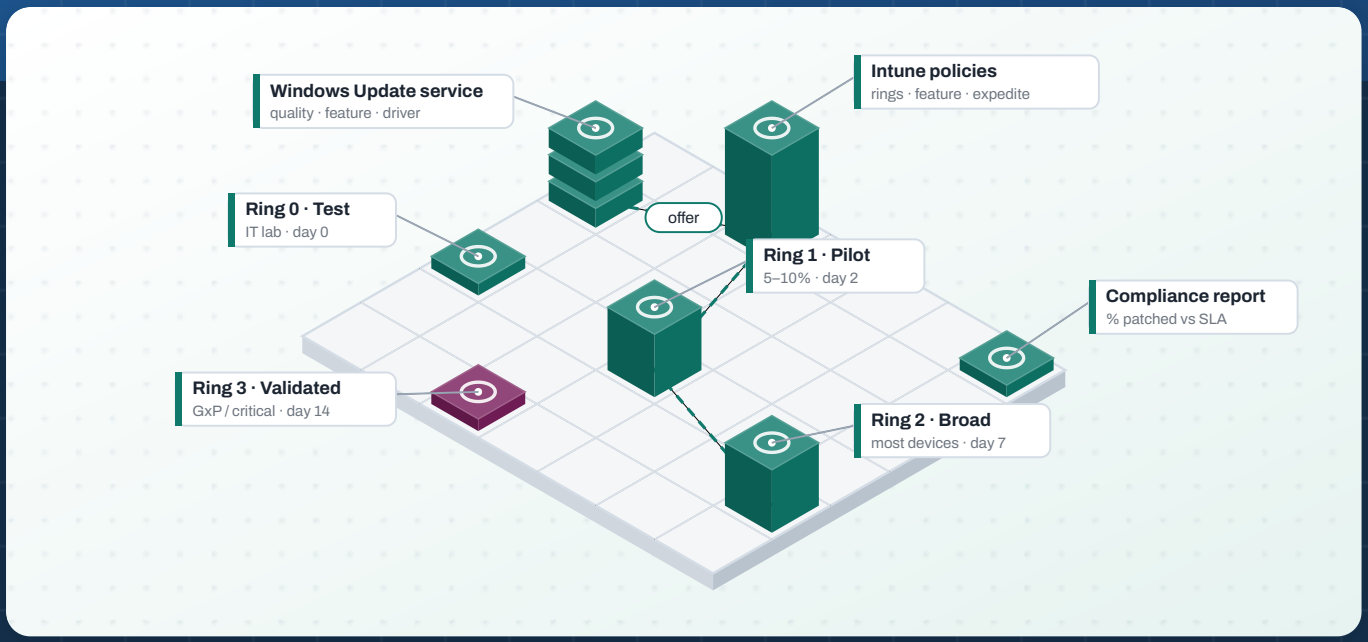
STANDARD OPERATING PROCEDURE · FIELD GUIDE SERIES

SOP-EM-005

Managing Windows Update Rings and Patch Compliance with Intune

Design deployment rings with Windows Update for Business, pin feature versions, expedite zero-day fixes, control drivers, and prove monthly patch compliance against an SLA, with a Patch Tuesday runbook, pause and rollback steps, and a troubleshooting path for devices that fall behind.

Endpoint Management | Windows Update for Business | Patch Compliance



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LEARNING OBJECTIVES

After working through this procedure you will be able to:

- Explain the four kinds of Windows update and which Intune policy controls each: quality, feature, driver and expedited.
- Design a ring model with deferrals, deadlines and grace periods that matches business risk, including a validated-systems ring.
- Build update rings, a feature update policy, an expedite profile and a driver policy in the Intune admin center, and report on them with Graph PowerShell.
- Run the monthly Patch Tuesday runbook and measure compliance against an SLA by severity in Intune and Log Analytics.
- Pause, roll back and troubleshoot: GPO and WSUS conflicts, stuck devices and the Windows Update log.

Time	Difficulty	Minimum access	Risk
½ day build · monthly 2–3 h runbook	Intermediate	Policy and Profile Manager (Intune)	Moderate: a bad update reaches every ring that is not paused

CHAPTER 1

Concepts You Need First

Windows devices already know how to download updates from Microsoft. What an organization needs is control over **when** each device takes them, **which version** of Windows it runs, and **evidence** that every device actually installed them. **Windows Update for Business** is that control layer, and Intune is where it is configured. This SOP does not use WSUS or Configuration Manager software updates; devices download directly from Microsoft, and policy decides the timing.

KEY CONCEPT · POLICY SETS TIMING, NOT CONTENT

An update ring does not choose updates. Microsoft publishes them; the ring tells each device how long to wait before it is offered them (**deferral**), how long the user may postpone the restart once installed (**deadline**), and how long a returning device gets before a forced restart (**grace period**). Deploying a patch therefore means managing time, and stopping a bad patch means **pausing** the rings that have not reached it yet.

1.1 The four kinds of update and what controls each

Windows Update delivers several kinds of content. Each is governed by a different Intune policy type, and confusing them is the most common reason a device does not do what the administrator expected.

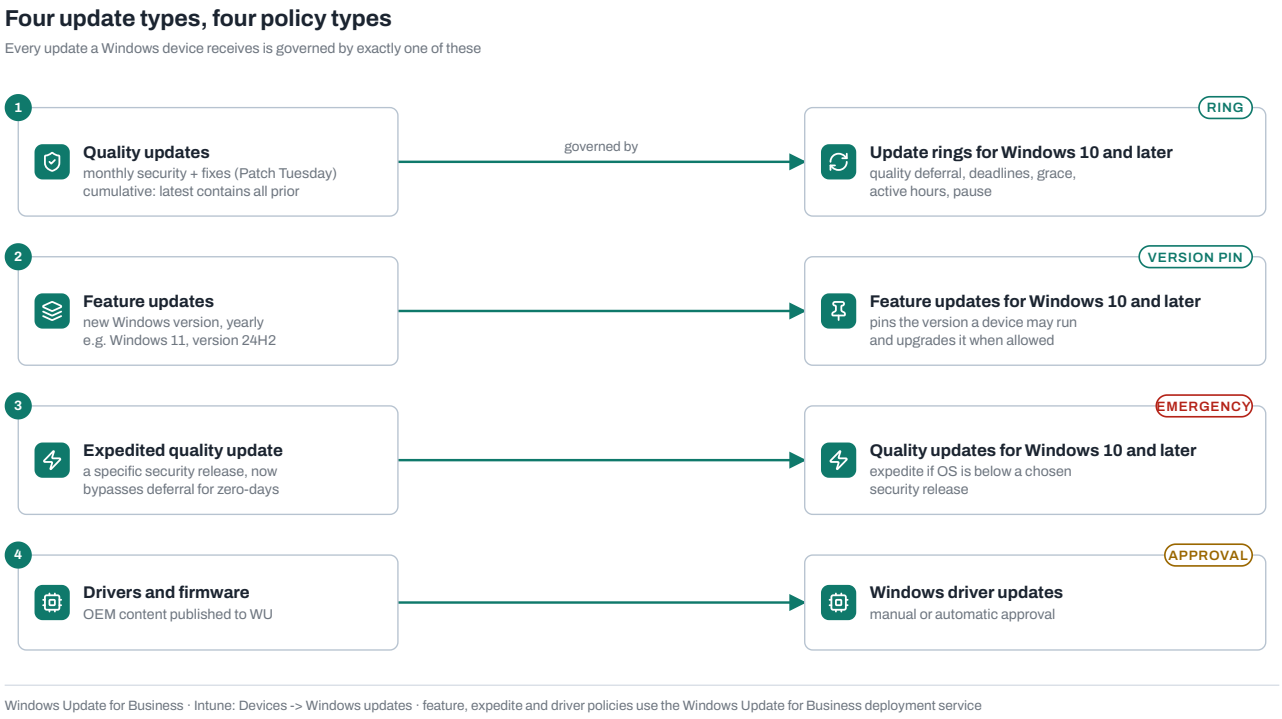


Figure 1. Each update type has its own policy. Rings never change the Windows version; feature update policies never schedule monthly patches.

Update type	Cadence	Size and impact	Intune policy
Quality (LCU)	Monthly, second Tuesday ("B" release); optional preview later in the month	Small to medium; one restart	Update ring: quality deferral 0–30 days
Out-of-band	As needed for urgent issues	Same as quality	Update ring, or an expedite profile
Feature	Annually (Windows 11 releases in the second half of the year)	Large; 30–90 min; one or more restarts	Feature update policy (pin version); rings can add feature deferral
Driver / firmware	As OEMs publish	Varies; firmware can be high risk	Driver update policy, or ring setting *Windows drivers: Allow/Block*
Microsoft product updates	Monthly	Office MSI, .NET, other Microsoft products	Update ring: *Microsoft product updates*

1.2 Deferral, deadline and grace period

Three numbers in an update ring decide when a device installs a quality update and when it restarts. Read them as a timeline that starts when Microsoft releases the update.

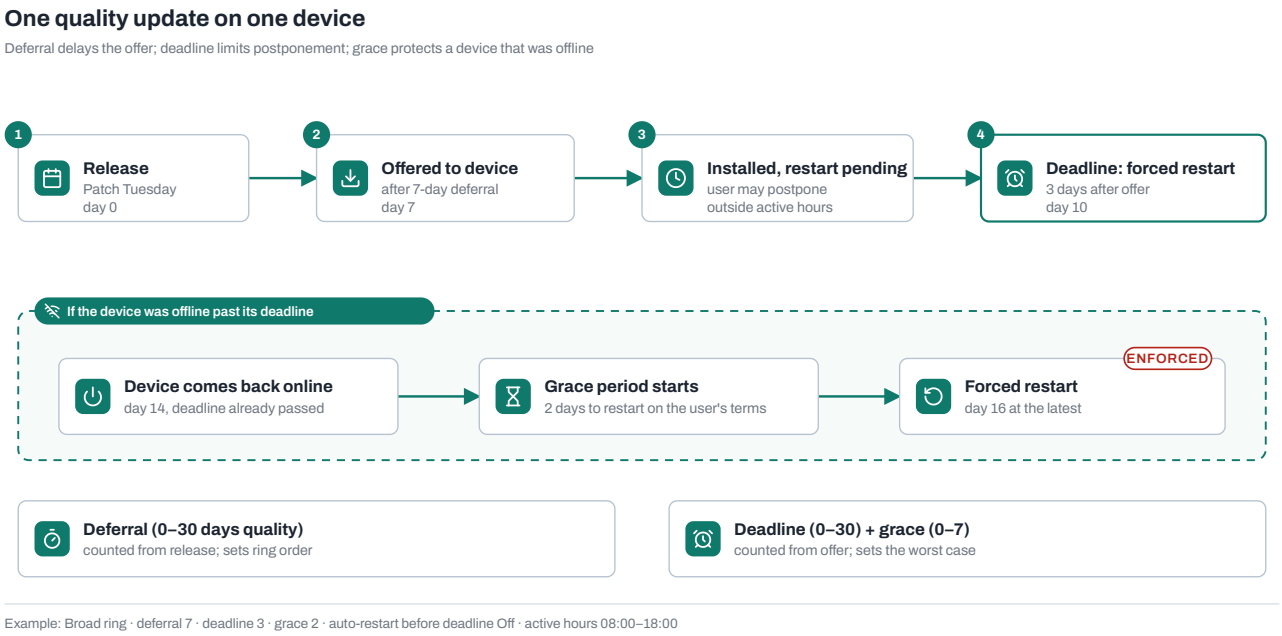


Figure 2. The worst case for a device is deferral + deadline + grace days after release. Ring design is choosing those three numbers per ring.

> **WATCH OUT · DEADLINES ARE WHAT MAKE COMPLIANCE PREDICTABLE**
>
> Without a deadline, a user can postpone a restart for weeks and the update stays *pending* in every report. Every ring in this SOP sets a quality deadline. The grace period exists only for devices that were off or offline, so keep it short.

1.3 Servicing channels and versions

Windows 11 devices in this SOP use the **General Availability Channel**: one feature update a year, supported for 24 months on Pro and 36 months on Enterprise and Education editions. Long-Term Servicing Channel (LTSC) editions are for fixed-function devices such as lab instruments and kiosks; they receive quality updates only and are covered by the validated ring. Windows Insider pre-release builds are never enabled outside a lab ring.

1.4 Windows Autopatch, and why this SOP still matters

Windows Autopatch is a Microsoft-operated service that creates and runs rings (Test, First, Fast, Broad) on the organization's behalf, with release management and reporting. It uses the same Windows Update for Business controls described here. Choose Autopatch when the team wants Microsoft to own the release cadence; follow this SOP when ring timing must be controlled locally, for example to align with change boards or validated systems. Do not run both against the same devices: Autopatch-registered devices must be excluded from the rings in this SOP.

1.5 Choosing a method

Method	Best for	Evidence produced
A · Intune admin center	Building and changing rings; monthly operations; pause and resume	Intune audit log, policy assignment status, screenshots
B · Microsoft Graph PowerShell	Exporting ring settings for review; reporting device state across rings; repeatable builds	JSON exports, CSV compliance reports
C · Log Analytics (WUfB reports)	Tenant-wide compliance, trends and alerting beyond 30 days	KQL query results, workbook snapshots

CHAPTER 2

Access, Prerequisites and Planning

2.1 Level of access required

Task	Minimum role	Notes
View rings, device update status, reports	Read Only Operator (Intune)	For reviews and audits.
Create and edit rings, feature, quality and driver policies	Policy and Profile Manager (Intune) or Intune Administrator	Scope tags limit which rings a delegated admin can change.
Pause, resume, uninstall a ring's updates	Policy and Profile Manager	Emergency action; record in the change ticket.
Approve drivers (manual approval policies)	Policy and Profile Manager	Driver approvals are a change; follow CM-3.
Enable Windows diagnostic data features	Intune Administrator	Tenant administration › Connectors and tokens › Windows data.
WUfB reports in Log Analytics	Log Analytics Contributor + Intune Administrator	Workspace in the same tenant.

2.2 Licensing and platform requirements

Update rings work on Windows 10 and 11 Pro, Enterprise and Education. Feature update policies, expedited quality updates and driver update policies use the **Windows Update for Business deployment service**, which needs a qualifying license such as Windows Enterprise E3/E5, Windows Education A3/A5 or Microsoft 365 Business Premium. Devices must be Microsoft Entra joined or hybrid joined and enrolled in Intune.

2.3 Prerequisites checklist

- [ ] Change ticket approved for the ring build, and a standing change for the monthly Patch Tuesday cycle.
- [ ] Windows diagnostic data enabled at tenant level: `Tenant administration` › `Connectors and tokens` › `Windows data` › *Enable features that require Windows diagnostic data in processor configuration* = **On**.
- [ ] Devices send at least **Required** diagnostic data (a settings catalog or device restriction policy sets *Allow Telemetry* to Required or higher).
- [ ] Inventory of policy conflicts: GPOs that set *Configure Automatic Updates*, *Specify intranet Microsoft update service location* (WSUS) or deferral settings, and any third-party patch agent.

- [ ] For co-managed devices: Configuration Manager co-management workload **Windows Update policies** moved to Intune for the pilot collection first.
- [ ] Ring membership groups created (dynamic or assigned): `WU-Ring0-Test`, `WU-Ring1-Pilot`, `WU-Ring2-Broad`, `WU-Ring3-Validated`.
- [ ] Devices registered with Windows Autopatch identified and excluded from these groups.
- [ ] Business owner sign-off for the validated ring (lab, manufacturing, GxP) and its longer timeline.
- [ ] Severity SLA agreed with vulnerability management (see section 2.5) and aligned with SOP-SEC-001.

2.4 Risk and planning notes

Action	Risk	Notes
Read reports, export ring settings	READ-ONLY	Always first.
Create rings and assign to Test ring only	LOW	Affects IT lab devices.
Assign rings to Pilot and Broad groups	MODERATE	Changes restart behaviour for most users; communicate first.
Remove conflicting GPO / WSUS settings	MODERATE	Devices switch update source; expect a scan spike and catch-up installs.
Pin or upgrade feature version	HIGH	Large download and long restart; app compatibility testing required.
Expedite a quality update	HIGH	Bypasses deferral for every targeted device; restarts within the chosen days.
Automatic driver / firmware approval	HIGH	Firmware failures can leave devices unbootable; use manual approval.

2.5 Decisions to make up front

Decision	Options	Baseline choice
Number of rings	2 to 5	Four: Test, Pilot, Broad, Validated
Ring membership	Assigned groups; dynamic by device attribute; random %	Assigned for Test, Pilot, Validated; Broad = all Windows devices minus the other three
Feature version	Latest; N-1; pinned	Pinned to Windows 11, version 24H2 until the next version passes Pilot
Driver updates	Allow in ring; block; driver policy	Block in rings; manual-approval driver policy for selected classes
Restart behaviour	Auto install and restart; notify; outside active hours	*Auto install and restart at maintenance time*, active hours 08:00–18:00

Decision	Options	Baseline choice
Patch SLA by severity	Org-defined	Actively exploited: 7 days · Critical: 14 · High: 30 · Medium: 60

CHAPTER 3

Designing the Ring Model

A ring is a group of devices that receives updates at the same time. Rings exist so that a bad update is seen on a few devices that IT can recover before it reaches everyone. Each ring is one *Update rings for Windows 10 and later* profile assigned to one group.

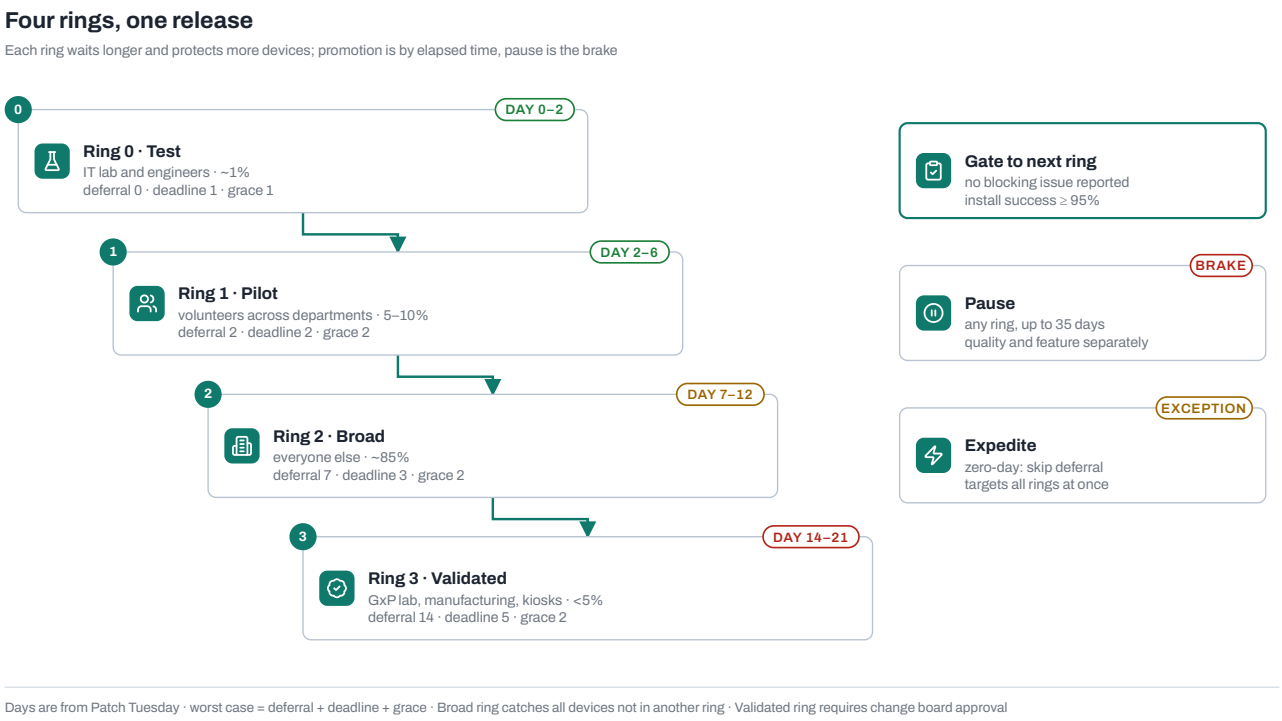


Figure 3. The baseline ring model. The validated ring trails the Broad ring by a week so that qualification can run on a known-good update.

3.1 Ring settings in detail

Setting	Ring 0 Test	Ring 1 Pilot	Ring 2 Broad	Ring 3 Validated
Servicing channel	General Availability	General Availability	General Availability	General Availability
Microsoft product updates	Allow	Allow	Allow	Allow
Windows drivers	Block	Block	Block	Block
Quality update deferral (days)	0	2	7	14
Feature update deferral (days)	0	0	0	0 (version pinned by feature policy)
Feature update uninstall period (days)	10	10	10	30

Setting	Ring 0 Test	Ring 1 Pilot	Ring 2 Broad	Ring 3 Validated
Pre-release builds	Not configured	Not configured	Not configured	Not configured
Automatic update behaviour	Auto install and restart	Auto install at maintenance time	Auto install at maintenance time	Auto install at maintenance time
Active hours	—	08:00–18:00	08:00–18:00	Per site shift pattern
Deadline for quality updates	1	2	3	5
Deadline for feature updates	2	5	7	14
Grace period	1	2	2	2
Auto restart before deadline	Yes	No	No	No

WHY · WHY FEATURE DEFERRAL STAYS AT ZERO

Feature update deferral in a ring and a feature update policy both control the Windows version, and mixing them makes the outcome hard to predict. This SOP sets ring feature deferral to 0 and lets the feature update policy alone decide the version. A device with no feature update policy and feature deferral 0 would be offered the newest version, so every Windows device must be in the scope of exactly one feature update policy.

3.2 The validated ring

Regulated systems (GxP lab workstations, instrument PCs, manufacturing HMIs) often require documented evidence that a change does not affect the validated state. The validated ring gives them a fixed, predictable window: the update has already run on the Broad ring for a week, the lab owner reviews the release and any known issues, and installation happens inside a scheduled maintenance window under an approved change. See SOP-LAB-003 for how these workstations are provisioned and SOP-GOV-001 for the change and evidence process. Instrument-controller PCs that cannot accept monthly changes are better served by an LTSC edition and a compensating control, not by an indefinitely paused ring.

CHAPTER 4

Method A · Building in the Intune Admin Center

Build and assign Ring 0 first, watch one Patch Tuesday cycle, then assign the remaining rings. Nothing in this chapter changes devices outside the Test group until step 6.

STEP 1 Clear the way: conflicts and diagnostic data

1. In Group Policy Management, find GPOs linked to computer OUs that configure *Windows Components* › *Windows Update*. For devices moving to Intune, unlink them or set the settings to **Not Configured**; in particular remove *Specify intranet Microsoft update service location*.
2. For co-managed devices, in Configuration Manager go to `Administration` › `Cloud Services` › `Cloud Attach` › `Properties` › `Workloads`, and move **Windows Update policies** to *Pilot Intune*, then *Intune*.
3. In Intune go to `Tenant administration` › `Connectors and tokens` › `Windows data`; turn on *Enable features that require Windows diagnostic data in processor configuration* and save.
4. Confirm the diagnostic data policy (settings catalog *System* › *Allow Telemetry* = Required) is assigned to all Windows devices.

STEP 2 Create the Ring 0 · Test update ring

1. Go to `Devices` › `Windows updates` › `Update rings` tab › `Create profile`.
2. Name `WU-Ring0-Test`; description with the deferral and deadline values and the change number.
3. `Update ring settings`: Microsoft product updates **Allow**; Windows drivers **Block**; Quality update deferral period **0**; Feature update deferral period **0**; Upgrade Windows 10 devices to latest Windows 11 release **No**; Set feature update uninstall period **10**.
4. `User experience settings`: Automatic update behavior *Auto install and restart*; Option to restart device outside of active hours as required; `Use deadline settings` **Allow**: Deadline for feature updates **2**, Deadline for quality updates **1**, Grace period **1**, Auto reboot before deadline **Yes**.
5. `Assignments` › Included groups `WU-Ring0-Test`. `Review + create`.

Repeat for Pilot, Broad and Validated with the values in section 3.1. Broad: include All devices is not used; include a dynamic group of Windows devices and exclude the other three ring groups.

STEP 3 Pin the Windows version with a feature update policy

1. `Devices` › `Windows updates` › `Feature updates` tab › `Create profile`.

2. Name `WU-Feature-Win11-24H2`. Feature deployment settings: *Feature update to deploy* **Windows 11, version 24H2**.
3. Rollout options: *Make update available as soon as possible* for Test; for the production profile choose *Make update available gradually* with first group availability, final group availability and days between groups, or a start date after the Pilot sign-off.
4. If devices are not eligible for Windows 11, decide whether to keep them on Windows 10 (*When a device isn't capable of running Windows 11, install the latest Windows 10 feature update*) and track them for replacement.
5. Assign to all Windows device groups. Every Windows device must be in exactly one feature update profile.

STEP 4 Create the emergency expedite profile (do not assign yet)

1. Devices › Windows updates › Quality updates tab › Create profile.
2. Name `WU-Expedite-Template`. Setting: *Expedite installation of quality updates if device OS version less than* › choose the security release to require.
3. *Number of days to wait before restart is enforced*: **1** (0–2 allowed).
4. Save **without assignments**. During a zero-day the profile is edited to the new release and assigned to the affected groups (section 6.5).

STEP 5 Create a driver update policy with manual approval

1. Devices › Windows updates › Driver updates tab › Create profile. Name `WU-Drivers-Manual`.
2. Approval method **Manually approve and deploy driver updates**.
3. Assign to the Test and Pilot groups first. Review the *Recommended drivers* list monthly; approve with an availability date per ring.
4. Firmware and BIOS updates are approved only after the OEM release notes are reviewed and a Test device has installed them.

STEP 6 Assign the production rings

1. After one Patch Tuesday cycle on Ring 0 with no unresolved issues, assign `WU-Ring1-Pilot`.
2. Two weeks later, assign `WU-Ring2-Broad` and `WU-Ring3-Validated` (Validated only under its own approved change).
3. Check Devices › Windows updates › select a ring › Device status: every assigned device should report *Succeeded* within 24 hours of check-in.

4. Export every profile (Method B) and attach the export to the change record as the version of record.

4.1 Tie compliance to Conditional Access

Rings deliver updates; compliance proves they were installed and lets Conditional Access act on devices that fall behind. In the Windows compliance policy (**Devices** › **Compliance** › **Create policy** › Windows 10 and later), set **Device Properties** › *Minimum OS version* to the build of the security release that is past its SLA, for example `10.0.26100.xxxx`. Set the non-compliance action to *Mark device noncompliant* after a grace period that matches the SLA. Devices below the build then fail CA006 in SOP-ID-001 and lose access to Exchange and SharePoint until they patch.

> **WATCH OUT · RAISE THE MINIMUM BUILD DELIBERATELY**
> Raise the minimum OS version only after the SLA has expired for that release and the Broad ring reports above 95 percent. Raising it on Patch Tuesday would mark most of the estate non-compliant on day 0.

CHAPTER 5

Method B · Microsoft Graph PowerShell

Graph is used here to export the configuration for review and to build reports that the portal does not provide in one view. Update rings are device configurations of type `windowsUpdateForBusinessConfiguration`; feature update policies are `windowsFeatureUpdateProfile` resources (beta endpoint).

5.1 Connect with least privilege

CONNECT · READ-ONLY SCOPES FOR REPORTING — POWERSHELL

```powershell
Connect-MgGraph -Scopes "DeviceManagementConfiguration.Read.All","DeviceManagementManagedDevices.Read.All" -
NoWelcome
Get-MgContext | Select-Object Account, TenantId, Scopes
```

5.2 Export the rings and feature policies

EXPORT UPDATE RINGS AND FEATURE UPDATE PROFILES TO JSON — POWERSHELL

```powershell
$stamp = Get-Date -Format yyyyMMdd-HHmm
$dir   = "C:\WU-Baseline\export-$stamp"; New-Item $dir -ItemType Directory | Out-Null

# Update rings
$rings = Get-MgDeviceManagementDeviceConfiguration -All |
  Where-Object { $_.AdditionalProperties.'@odata.type' -eq
'#microsoft.graph.windowsUpdateForBusinessConfiguration' }
foreach ($r in $rings) {
  $full = Invoke-MgGraphRequest -Method GET -Uri "v1.0/deviceManagement/deviceConfigurations/$($r.Id)"
  $full | ConvertTo-Json -Depth 10 | Out-File "$dir\ring-$($r.DisplayName).json" -Encoding utf8
}

# Feature update profiles (beta)
$fu = Invoke-MgGraphRequest -Method GET -Uri "beta/deviceManagement/windowsFeatureUpdateProfiles"
$fu.value | ForEach-Object {
  $_ | ConvertTo-Json -Depth 10 | Out-File "$dir\feature-$($_.displayName).json" -Encoding utf8 }
```

5.3 Summarise ring settings for review

ONE ROW PER RING: DEFERRALS, DEADLINES, PAUSE STATE — POWERSHELL

```powershell
$rings | ForEach-Object {
  $p = (Invoke-MgGraphRequest -Method GET -Uri "v1.0/deviceManagement/deviceConfigurations/$($_.Id)")
  [pscustomobject]@{
    Ring            = $p.displayName
    QualityDeferral = $p.qualityUpdatesDeferralPeriodInDays
    FeatureDeferral = $p.featureUpdatesDeferralPeriodInDays
    QualityDeadline = $p.deadlineForQualityUpdatesInDays
    FeatureDeadline = $p.deadlineForFeatureUpdatesInDays
    Grace           = $p.deadlineGracePeriodInDays
    QualityPaused   = $p.qualityUpdatesPaused
    FeaturePaused   = $p.featureUpdatesPaused
    Drivers         = if ($p.driversExcluded) { 'Blocked' } else { 'Allowed' }
  }
} | Format-Table -AutoSize
```

5.4 Report devices below the required build

WINDOWS DEVICES BELOW A TARGET OS BUILD, WITH LAST CHECK-IN — POWERSHELL

```powershell
$target = [version]'10.0.26100.4061'   # example: build of the release past SLA
Get-MgDeviceManagementManagedDevice -All -Filter "operatingSystem eq 'Windows'" `
  -Property deviceName,osVersion,lastSyncDateTime,userPrincipalName,complianceState |
  Where-Object { [version]$_.OsVersion -lt $target } |
  Select-Object DeviceName, OsVersion, LastSyncDateTime, UserPrincipalName, ComplianceState |
  Sort-Object LastSyncDateTime |
  Export-Csv ".\below-target-$(Get-Date -Format yyyyMMdd).csv" -NoTypeInformation
```

NOTE · CHANGE THROUGH THE PORTAL, REPORT THROUGH GRAPH

Ring changes are rare and benefit from the portal's validation. If ring changes are scripted, use the export as the body, change only the intended properties, remove read-only properties (`id` , `createdDateTime` , `lastModifiedDateTime` , `version`) and send a PATCH with `Invoke-MgGraphRequest` , under the scope `DeviceManagementConfiguration.ReadWrite.All` granted for the change window only.

CHAPTER 6

Operating the Monthly Cycle

Patch Tuesday is the second Tuesday of each month, when Microsoft releases security updates at about 10:00 Pacific time. The runbook below turns that release into a measured, evidenced outcome within the SLA.

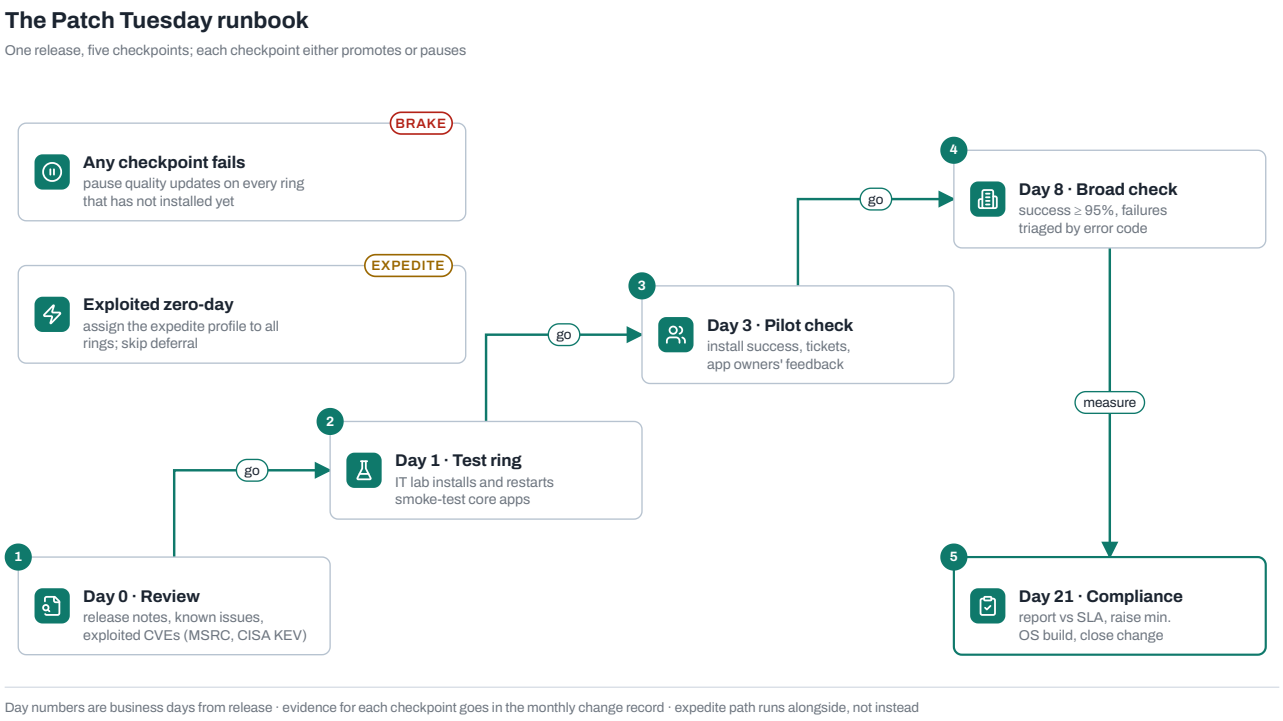


Figure 4. Promotion is automatic by deferral; the runbook's job is to check each ring in time to pause the next one if needed.

6.1 Checkpoint tasks

Checkpoint	Task	Evidence
Day 0 · Review	Read the Windows release health dashboard and release notes; list CVEs marked *Exploitation Detected*; check the CISA Known Exploited Vulnerabilities catalogue; decide whether to expedite	Release summary in the change record; expedite decision
Day 1 · Test	Confirm Test devices installed and restarted; open core apps (Office, VPN, line-of-business, label printing)	Ring device status export; smoke-test checklist
Day 3 · Pilot	Install success rate; service desk tickets tagged *patch*; app owner sign-off for critical apps	Pilot report; ticket query
Day 8 · Broad	Success rate ≥ 95%; top failure codes triaged; offline devices listed	Windows update report export

Checkpoint	Task	Evidence
Day 14 · Validated	Lab owner approval under change; install in maintenance window; post-install verification	Change ticket with objective evidence
Day 21 · Compliance	Compliance against SLA; raise compliance minimum OS build; exceptions recorded	Monthly compliance report; exception register

6.2 Reporting in Intune

Reports › Windows updates gives a summary and per-policy reports. The **Windows Feature Update** report shows each device's progress toward the pinned version; the **Windows Expedited Update** report shows expedite progress; the driver report shows approved drivers by state. For monthly quality updates, use each ring's Device status and the *Windows update distribution* and compliance reports. Intune reports are near-real-time but cover limited history.

6.3 Reporting in Log Analytics (Windows Update for Business reports)

Windows Update for Business reports sends device update data to a Log Analytics workspace (tables prefixed `UC`), giving history, cross-tenant views and alerting. Enable it from the Azure portal (Monitor › Workbooks › Windows Update for Business reports) and point it at the workspace used for SOP-SEC-001. Data appears within about 24–48 hours of enrollment.

DEVICES NOT ON THE LATEST SECURITY UPDATE, BY OS BUILD — KQL

```
UCClient
| where TimeGenerated > ago(1d)
| summarize arg_max(TimeGenerated, *) by AzureADDeviceId
| summarize Devices = count() by OSSecurityUpdateStatus, OSVersion, OSBuild
| order by Devices desc
```

QUALITY UPDATE INSTALL STATE FOR THIS MONTH, PER STATE — KQL

```
UCClientUpdateStatus
| where TimeGenerated > ago(2d)
| where UpdateCategory == "WindowsQualityUpdate"
| summarize arg_max(TimeGenerated, *) by AzureADDeviceId
| summarize Devices = count() by ClientState, ClientSubstate
| order by Devices desc
```

NOTE · SCHEMA CHECK

Column names in the `UC*` tables are documented by Microsoft and occasionally extended. Run `UCClient | getschema` once when building the workbook, and keep the queries in the change repository next to the ring exports.

6.4 The compliance KPI

Report one number to management each month: the percentage of active Windows devices (checked in within 14 days) that have installed the security release now past its SLA. Report it per ring and per severity class.

Severity of the release's worst CVE	SLA (days from release)	Target	Escalation if missed
Actively exploited (any CVSS)	7	≥ 98% of active devices	Expedite; CISO notified
Critical (CVSS 9.0–10.0)	14	≥ 98%	Device list to owners; compliance gate
High (7.0–8.9)	30	≥ 95%	Monthly report; exception required
Medium (4.0–6.9)	60	≥ 95%	Quarterly review

6.5 Expediting a zero-day fix

STEP 1 **Expedite an exploited vulnerability**

1. Confirm the CVE is exploited and fixed in a specific release (MSRC *Exploitation Detected*, CISA KEV, or vendor advisory). Record the decision in an emergency change.
2. Open `WU-Expedite-Template` › edit *Expedite installation of quality updates if device OS version less than* › select the release that contains the fix. Restart enforcement **1** day.
3. Assign to `WU-Ring0-Test` first; after 2–4 hours with successful installs, add Pilot, Broad and Validated (Validated only with the lab owner's approval).
4. Monitor **Reports** › **Windows updates** › **Reports** › *Windows Expedited Update Report*.
5. When the release is installed above target, remove the assignments. Expedite profiles have no effect on devices already at or above the chosen build.

Expedite requires Windows diagnostic data enabled and the deployment service license; devices that are off will install when they next come online.

CHAPTER 7

Verification, Rollback and Troubleshooting

7.1 Verification checklist

Check	How	Expected
Every Windows device in exactly one ring	Group membership query; ring device status counts	Sum of ring devices = Windows device count (minus Autopatch)
Every device in one feature update profile	Feature update report	No device shows *Not in scope*
No GPO or WSUS conflict	On a sample device, run the registry check in section 7.3 (Policies key for GPO, PolicyManager key for MDM)	No `WUServer` value under the Policies key; ring values present under the PolicyManager *Update* key
Settings applied on device	Settings › Windows Update › *Some update policies are managed by your organization* › view policies	Deferral and deadline match the ring
Diagnostic data flowing	Feature and expedite reports populated	Devices listed within 48 hours
Monthly KPI	Log Analytics query / Intune report	Targets in section 6.4 met per ring
Evidence	Change repository	Ring and feature exports dated this change

7.2 Pause and rollback

Situation	Action	Notes
Bad quality update reported in Test or Pilot	Open each later ring › **Pause** › *Quality updates*	Pause lasts 35 days unless extended or resumed; devices that already installed are unaffected.
Bad quality update already installed	Ring › **Uninstall** › *Quality updates* (removes the latest quality update), then pause	Works only for the most recent quality update and while it is still installed; confirm the KB is removable.
Feature update breaks an app	Ring › **Uninstall** › *Feature updates*, within the uninstall period	Rolls back to the previous version; then edit the feature profile to keep the older version.
Faulty driver or firmware	Driver policy › pause or decline the driver	Already-installed drivers are rolled back on the device (Device Manager › Roll Back Driver) or by OEM tooling.

Situation	Action	Notes
Resume	Ring › `Resume` after the fix is published	Record pause and resume times in the change record.

WARNING · PAUSE IS TEMPORARY BY DESIGN

A paused ring automatically resumes after 35 days, and a long pause silently pushes those devices past their SLA. Every pause has an owner and a review date in the change record, and the monthly report lists paused rings.

7.3 Troubleshooting

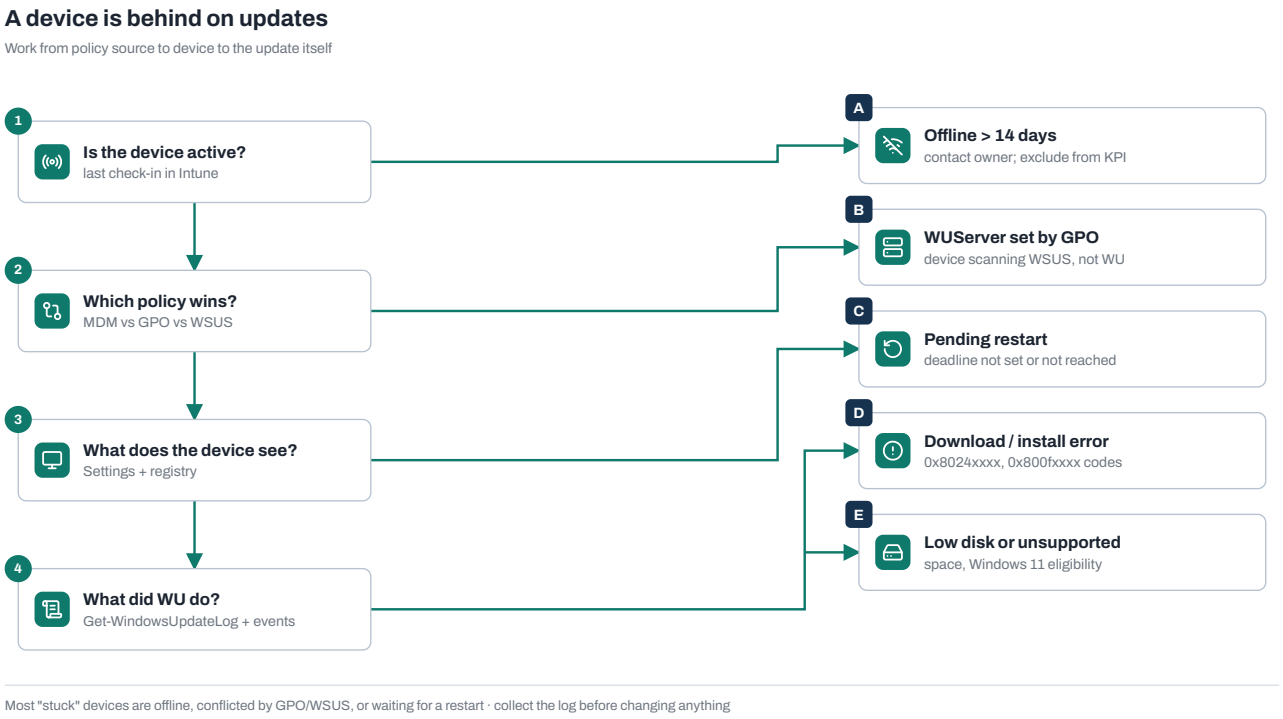


Figure 5. Troubleshooting runs from the policy source down to the Windows Update client. Each question rules out one family of causes.

ON THE DEVICE · WHAT POLICY AND WHAT STATE — POWERSHELL

```powershell
# Policy delivered by Intune (MDM)
Get-ItemProperty 'HKLM:\SOFTWARE\Microsoft\PolicyManager\current\device\Update' |
  Select-Object DeferQualityUpdatesPeriodInDays, ConfigureDeadlineForQualityUpdates,
                ConfigureDeadlineGracePeriod, PauseQualityUpdates, ExcludeWUDriversInQualityUpdate

# Group Policy / WSUS settings that may conflict
Get-ItemProperty 'HKLM:\SOFTWARE\Policies\Microsoft\Windows\WindowsUpdate' -ErrorAction SilentlyContinue |
  Select-Object WUServer, WUStatusServer, DisableDualScan, SetPolicyDrivenUpdateSourceForQualityUpdates

# Installed updates, newest first, and pending reboot
Get-HotFix | Sort-Object InstalledOn -Descending | Select-Object -First 5
Test-Path 'HKLM:\SOFTWARE\Microsoft\Windows\CurrentVersion\WindowsUpdate\Auto Update\RebootRequired'

# Build the readable Windows Update log from ETW traces (writes WindowsUpdate.log to the desktop)
Get-WindowsUpdateLog
Get-WinEvent -LogName 'Microsoft-Windows-WindowsUpdateClient/Operational' -MaxEvents 30 |
  Select-Object TimeCreated, Id, LevelDisplayName, Message
```

Symptom	Likely cause	Fix
Device reports *Succeeded* for the ring but has not installed last month's update	A GPO points it at WSUS (`WUServer`), so it scans WSUS	Remove the WSUS GPO settings for Intune-managed devices; `gpupdate /force`; rescan.
Settings page shows options greyed and different deferrals	Both GPO and MDM configure the same setting	Remove the GPO setting; for co-managed devices check the Windows Update policies workload is on Intune.
Update stuck at *Pending restart* for weeks	Ring has no quality deadline, or the user is always active	Set *Use deadline settings*; verify the deadline value reached the device.
Feature update never offered	Device not in a feature update profile, ineligible hardware, or Safeguard hold	Check the feature update report for the device's alert; safeguard holds clear when Microsoft resolves the issue.
Expedite shows *Not applicable*	Device already at or above the chosen build, or diagnostic data off	Confirm the build; confirm the Windows data connector and telemetry level.
Error 0x80070070	Not enough disk space	Free space (Storage Sense, clean-up script) and retry; see SOP-WIN-004 for disk commands.

CHAPTER 8

Summary, Glossary and Review

8.1 Chapter summary

- Quality, feature, expedited and driver updates are controlled by four different Intune policy types.
- A ring sets timing: deferral delays the offer, deadline limits postponement, grace protects returning devices.
- Four rings (Test, Pilot, Broad, Validated) with increasing deferral give time to find and pause a bad update.
- Feature versions are pinned by one feature update policy per device; ring feature deferral stays at zero.
- GPO and WSUS settings must be removed for Intune-managed devices, or devices will not follow the rings.
- Monthly compliance is measured against an SLA by severity, enforced through the compliance minimum OS build and Conditional Access.
- Pause stops later rings; uninstall removes the latest update; both are recorded in the change.

Check your understanding

1. A device is in a ring with deferral 7, deadline 3 and grace 2, and is switched off from day 5 to day 15. When is its forced restart?
2. Why does this SOP set feature update deferral to zero in every ring?
3. An exploited vulnerability is fixed in this month's release. Which profile do you use, and how does it interact with the ring deferrals?
4. A device shows the right ring settings in Intune but still scans an on-premises server. Which registry value would you check, and what is the fix?
5. When should the compliance policy's minimum OS version be raised, and what happens to devices below it?

8.2 Glossary

Windows Update for Business	Set of Windows policies that control when devices take updates from Microsoft's service.
Quality update (LCU)	Monthly cumulative security and reliability update.
Feature update	Upgrade to a new Windows version, released annually.
Deferral	Days after release before a device is offered an update.
Deadline	Days after the offer before installation and restart are forced.

Grace period	Minimum days a device that missed its deadline gets before a forced restart.
Ring	A group of devices with the same update timing; one update ring profile.
Expedite	Deployment service feature that installs a specified security release immediately, ignoring deferral.
Safeguard hold	Microsoft block on a feature update for devices with a known compatibility issue.
Dual scan	Condition where a device configured for WSUS also scans Windows Update; a source of conflicts.
Windows Autopatch	Microsoft-managed service that operates update rings on the customer's behalf.

8.3 Framework references

Framework	Controls addressed
NIST SP 800-53 Rev. 5	SI-2 Flaw Remediation · SI-2(2) Automated Flaw Remediation Status · SI-2(3) Time to Remediate · CM-3 Configuration Change Control · CM-8 System Component Inventory · RA-5 Vulnerability Monitoring and Scanning
CIS Controls v8	7.3 Perform automated operating system patch management · 7.4 Perform automated application patch management · 7.7 Remediate detected vulnerabilities
CIS Microsoft Intune for Windows Benchmark	Windows Update section: deferral, deadline and pre-release settings
FDA 21 CFR Part 11 / GAMP 5	Validated ring with documented change control and objective evidence for regulated systems (§11.10(k))